

Exam Date & Time: 25-Sep-2025 (11:30 AM - 01:00 PM)



MANIPAL UNIVERSITY JAIPUR

Odd Semester Mid Term Examination, September 2025
Faculty of Science, Technology & Architecture, School of Engineering
Department of Electronics & Communication Engineering
B.Tech. (Electronics & Communication Engineering)

All questions are compulsory.
Missing data, if any, may be assumed suitably.

Digital Electronics [ECE2102]

Marks: 30

Duration: 90 mins.

SECTION A (Memory Based Questions)

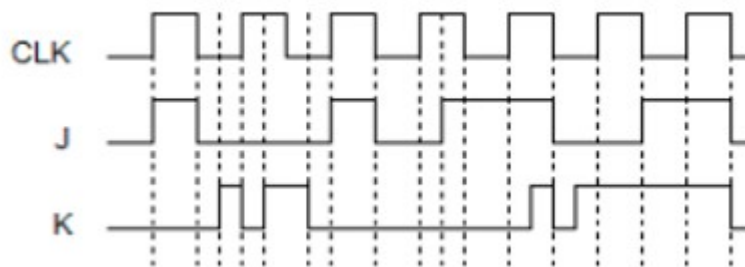
Answer all the questions.

- 1) Optimized the following expression and implement using 2-input NOR gate.
 $F = P'Q' + P'$ (2)
- 2) The four-bit binary sequence 1011 is applied to an even parity generator circuit. What will be the output? (2)
- 3) What excitation input should be applied to a T- FF to make output toggle? (2)

SECTION B (Concept Based Questions)

Answer all the questions.

- 4) Minimize the following Boolean function using K-Map method to obtain a simplified expression. Also draw the logic circuit using basic gates.
 $F(a,b,c,d) = \sum m(0, 4, 5, 7-9, 15) + \sum d(1, 3, 6, 14)$ (4)
- 5) Implement the following function using single 8:1 Multiplexer.
 $F(P,Q,R,S) = \sum m(0, 1, 3, 4, 8, 9, 10, 11, 13, 15)$ (4)
- 6) Design Full Adder using suitable decoder. (4)
- 7) Draw the output waveform for the following input waveform, assuming the flip-flop to be positive edge triggered and the output Q is initially at logic zero. (4)



SECTION-C (Analytical Based Questions)

Answer all the questions.

- 8) Design a synchronous counter that counts clock pulses from 0-7 using negative edge triggered T flipflop. (8)

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